

Name: _____

Date: _____

Unit 4: Chemical Dynamics and Equilibrium Unit TestKnowledge (20 marks) – Multiple Choice on D2L

sig fig

Write pH values to 2 decimal places for all pH calculations. Write final answers to the correct number of significant figures for all other types of calculations.

Thinking (16 marks)

③

1. Calculate the pH of a 1.00 L solution prepared by dissolving 27.0 g sodium hydroxide, NaOH(aq), in water. (4 marks)

$$n_{\text{NaOH}} = \frac{27.0 \text{ g}}{(22.99 \text{ g/mol} + 16.00 \text{ g/mol} + 1.01 \text{ g/mol})}$$

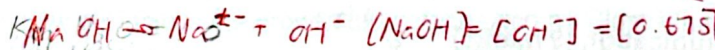
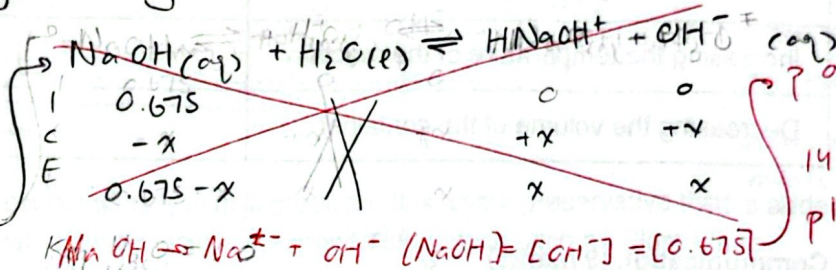
$$= 0.675 \text{ mol}$$

$$c = \frac{0.675 \text{ mol}}{1.00 \text{ L}}$$

$$= 0.675 \text{ mol/L}$$

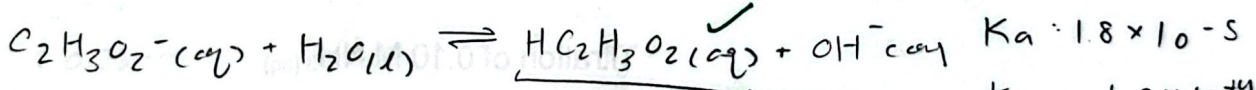
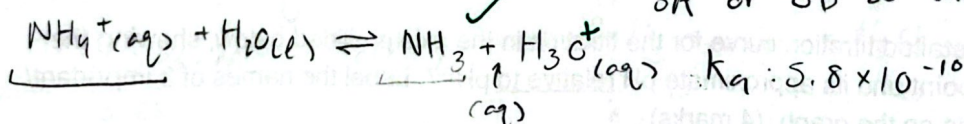
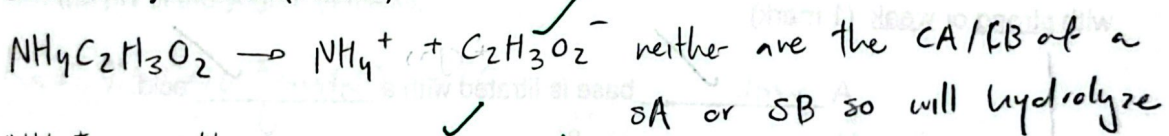
$$\text{pH} = 14 - (-\log([\text{OH}^-]))$$

NO ICE Table for strong Base Acid!



$$\begin{aligned} \text{pOH} &= -\log(0.675) \\ &= 0.17 \\ \text{pH} &= 14 - 0.17 = 13.83 \end{aligned}$$

2. Determine whether a solution of ammonium acetate ($\text{NH}_4\text{C}_2\text{H}_3\text{O}_2$) is acidic, basic, or neutral. Show all of your work! (4 marks)



$$K_b = \frac{1.0 \times 10^{-14}}{1.8 \times 10^{-5}} = 5.5 \times 10^{-10}$$

$K_a > K_b$
∴ the solution is acidic

3. What is the pH of a solution when 20 mL of 0.20 mol/L NaOH is titrated with 10 mL of 0.10 mol/L HCl? (4 marks)

$$c = \frac{n}{V}$$

$$n = cV$$

$$\begin{aligned} n_{\text{NaOH}} &= (0.02 \text{ L})(0.20 \text{ mol/L}) \\ &= 4 \times 10^{-3} \text{ mol} \leftarrow [\text{OH}^-] \text{ bc 1:1 mol ratio} \end{aligned}$$

$$\begin{aligned} n_{\text{HCl}} &= (0.01 \text{ L})(0.10 \text{ mol/L}) \\ &= 1 \times 10^{-3} \text{ mol} \leftarrow [\text{H}^+] \text{ bc 1:1 mol ratio} \end{aligned}$$

$$4 \times 10^{-3} - 1 \times 10^{-3} = 3 \times 10^{-3} \text{ mol OH}^-$$

$$\begin{aligned} \text{pOH} &= -\log(3 \times 10^{-3}) \\ &= 2.5228 \end{aligned}$$

$$\text{pH} = 14 - 2.5228$$

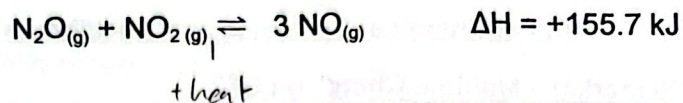
$$= 11.48 \quad \boxed{\text{pH} = 11.48}$$

$$\boxed{\text{pH} = 11.48}$$

$$c = \frac{0.003 \text{ mol}}{0.030 \text{ L}} \leftarrow \text{total volume}$$

SCH 4U1a

4. Which direction will the following equilibrium shift when exposed to the changes below? (3 marks)



Stress	Direction of Shift
Addition of N_2O	$\rightarrow$ ✓
Addition of NO	$\leftarrow$ ✓
Addition of Ne to the container	n/a ✓
Removal of NO_2	$\leftarrow$ ✓
Increasing the temperature of the reaction	$\rightarrow$ ✓
Decreasing the volume of the container	$\leftarrow$ ✓

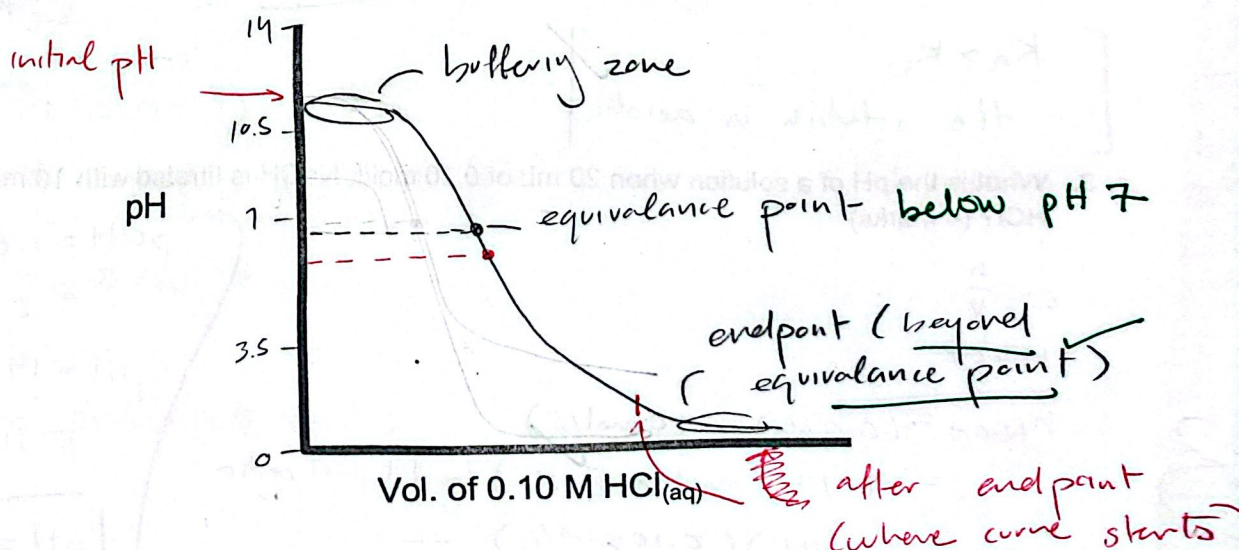
Communication (9 marks)

1. a) You are given a solution of $\text{NH}_3(\text{aq})$ that has a concentration of 0.10 M. The solution was titrated with 0.10 M $\text{HCl}(\text{aq})$. Identify the strengths of acid and the base by filling in the following blanks with strong or weak: (1 mark)

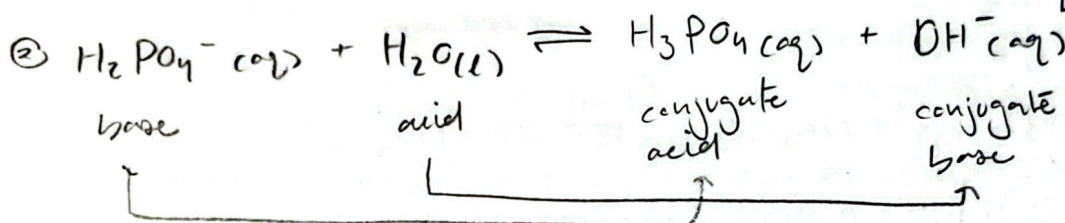
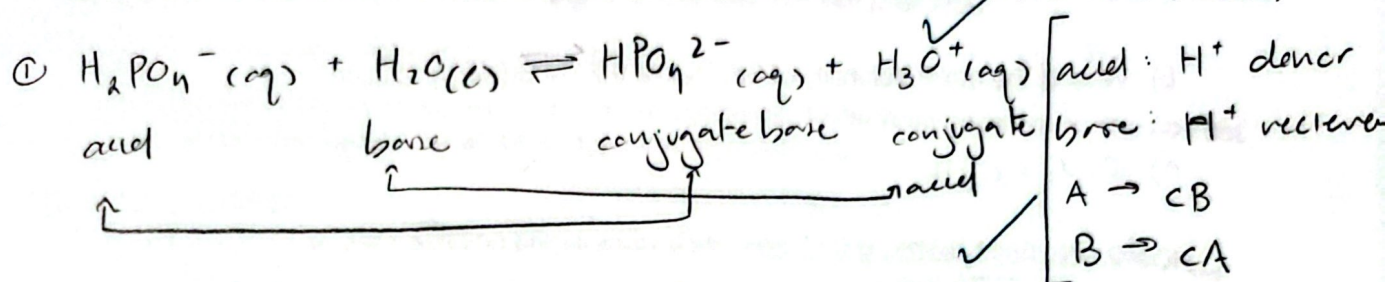
A weak base is titrated with a strong acid.

- b) Sketch a detailed titration curve for the titration in the axis provided below, showing the equivalence point and its approximate pH relative to $\text{pH}=7$. Label the names of 3 important/ distinct regions on the graph. (4 marks)

Titration of 0.10 M $\text{NH}_3(\text{aq})$



2. Show, using equations, how according to the Bronsted-Lowry theory, the borate ion, H_2PO_4^- (aq) is amphiprotic/amphoteric. Label the acid, base, conjugate acid, and conjugate base terms and identify the conjugate acid-base pairs in each reaction. Make sure to show all your work. (4 marks)



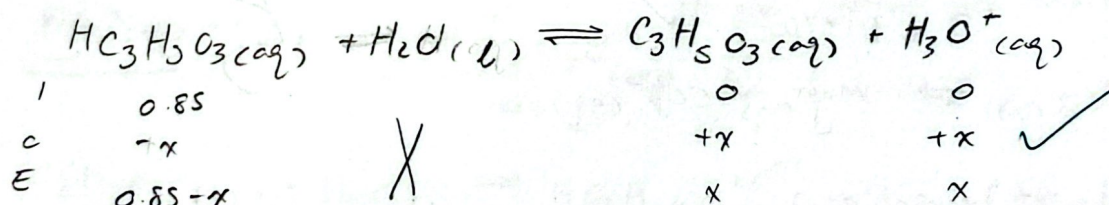
H_2PO_4^-
can act
as both
base and
acid

Application (10 marks)

1. Lactic acid, $\text{HC}_3\text{H}_5\text{O}_3$ (aq), occurs naturally in milk products. It is also a preservative that is added to foods to keep the pH down. Low pH inhibits the growth of bacteria such as *Clostridium botulinum*, the bacteria responsible for botulism.

- (a) Lactic acid ($K_a = 7.94 \times 10^{-5}$) is found in yogurt at a concentration of 0.85 mol/L. Determine the pH of the yogurt. (4 marks)

$$K_a = 7.94 \times 10^{-5} \quad 0.85 \text{ mol/L}$$



$$K_a = \frac{(C_3H_5O_3)(H_3O^+)}{(HC_3H_5O_3)}$$

100 rule: $\frac{0.85}{7.79 \times 10^{-5}} > 100$

$$7.99 \times 10^{-5} = \frac{x^2}{0.85}$$

$$\sqrt{(7.94 \times 10^{-5})(0.85)} = x$$
$$x = 8.21 \times 10^{-3} \leftarrow [\text{H}_3\text{O}^+] \text{ bc } 1:1 \text{ mol ratio}$$
$$\text{pH} = -\log(8.21 \times 10^{-3})$$
$$= 2.09$$
$$\boxed{\text{pH} = 2.09}$$

- (b) *Clostridium botulinum* will grow in foods with a pH of 4.8 to 7. Is this yogurt safe to eat? Give a brief explanation. (1 mark)

the yogurt mentioned above is safe to eat with a pH of

2.09, so *Clasidium botulinum* will not grow in it as it only grows in foods with pH 4.6-7

2. Choose only ONE (1) of the following to complete: (5 marks)

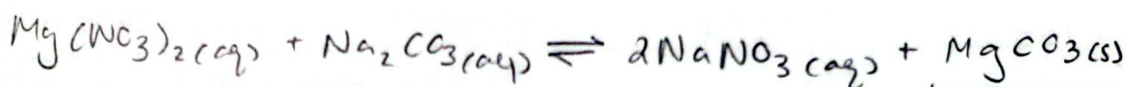
- a) Will a precipitate form by mixing 25.0 mL of 0.0010 mol/L $\text{Mg}(\text{NO}_3)_2$ with 50.0 mL of 0.010 mol/L Na_2CO_3 ?

OR

- b) What is the molar solubility of SrCO_3 in a 0.20 M $\text{Sr}(\text{NO}_3)_2$ solution?

a) $C_1V_1 = C_2V_2$

$$C_2 = \frac{C_1V_1}{V_2}$$

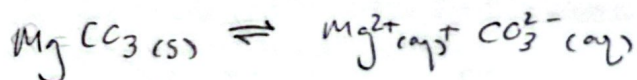


$$[\text{Mg}(\text{NO}_3)_2] = \frac{(0.025\text{L})(0.0010\text{mol/L})}{(0.025 + 0.050\text{L})} \quad 1:1 \text{ mol ratio}$$

$$= 3.3 \times 10^{-4} \text{ mol/L}$$

$$[\text{Na}_2\text{CO}_3] = \frac{(0.050\text{L})(0.010\text{mol/L})}{(0.050\text{L} + 0.025\text{L})}$$

$$= 6.6 \times 10^{-3}$$



$$Q = \frac{[\text{Mg}^{2+}][\text{CO}_3^{2-}]}{[\text{MgCO}_3]}$$

$$Q = (6.6 \times 10^{-3})(3.3 \times 10^{-4})$$

$$Q = 2.178 \times 10^{-6} \rightarrow \boxed{2.2 \times 10^{-6}}$$

$$K_{sp} \text{ MgCO}_3 = 6.8 \times 10^{-6}$$

$Q < K_{sp} \therefore$ no precipitate forms